1 Preliminaries and Problem Setting

We begin by reviewing relevant background and formalizing the problem setting.

1.1 Regularized Quasi-Newton Method

The regularized quasi-Newton method is a variant of quasi-Newton methods that does not rely heavily on line search techniques. This property is important for optimization in noisy evaluation environments. There has been renewed interest in regularized Newton-type methods, including cubic-regularization schemes [30, 34, 42], and we focus on quadratic regularization [8, 22, 24, 36, 39, 40, 47] for simplicity of analysis and implementation.

For $g_k := \nabla f(x_k)$ and an approximate Hessian $B_k \in \mathbb{R}^{n\times n}$, regularized quasi-Newton methods compute the step as

$$x_{k+1} = x_k + d_k, \quad d_k = -(B_k + \mu_k I)^{-1} g_k,$$

with a regularization parameter $\mu_k \geq 0$. Combining regularization with line search is also possible and has been explored [39]. Several strategies for choosing μ_k in nonconvex optimization has been proposed [36, 39, 40], and one of them is given by

$$\mu_k = c_1 \max(0, -\lambda_{\min}(B_k)) + c_2 \|\nabla f(x_k)\|^{\delta},$$

where $c_1 > 1$, $c_2 > 0$, $\delta \geq 0$, and $\lambda_{\min}(B_k)$ denotes the smallest eigenvalue of B_k. This choice guarantees that $B_k + \mu_k I$ is positive definite, ensuring that d_k is a descent direction. We adapt such regularization strategies to noisy evaluation environments.

1.2 Objective-Function-Free Optimization

Objective-Function-Free Optimization (OFFO) refers to a class of methods that avoid explicit evaluations of the objective function and rely solely on gradient information [17, 18]. Such methods are particularly attractive when function values are unreliable due to noise. Similar adaptive trust-region approaches are also discussed [15].

Prominent examples of OFFO include adaptive gradient methods such as AdaGrad [10] and AdaGrad-Norm [43]. Their update rules are

$$\text{(AdaGrad)} \quad x_{k+1}^{(i)} = x_k^{(i)} - \frac{1}{\theta\sqrt{\varsigma + \sum_{j=0}^{k} (g_j^{(i)})^2}} g_k^{(i)}, \qquad (i = 1, 2, \ldots, n)$$

$$\text{(AdaGrad-Norm)} \quad x_{k+1} = x_k - \frac{1}{\theta\sqrt{\varsigma + \sum_{j=0}^{k} \|g_j\|^2}} g_k$$

respectively, where $\varsigma > 0$ and $\theta > 0$ are algorithmic parameters, and $x^{(i)}$ denotes the i-th component of a vector x.

1.3 Problem Setting

We consider optimization problems in which exact objective function values are unavailable, following the recent literature on error-aware optimization [2, 5, 6, 16, 35, 38]. When both objective values and gradients are heavily corrupted by noise, reliable optimization cannot be guaranteed. Thus, meaningful optimization requires that at least one source of information be sufficiently accurate. This work focuses on problems arising from finite-precision computations or inexact evaluations, where objective function evaluations are subject to non-negligible errors while gradient information can be computed with relatively high accuracy. Settings in which objective values are accurate but gradients are noisy, such as derivative-free optimization or stochastic approximation methods [3, 37, 44], are beyond the scope of this work.

In the following, we formalize the assumptions on function value and gradient evaluations. We assume that only an error-contaminated function value $\overline{f}(x)$ is accessible, satisfying the hybrid absolute-relative error model [20, 21]:

$$\left|f(x)-\overline{f}(x)\right| \leq \varepsilon_{\mathrm{f}} \max(1,|f(x)|), \tag{1}$$

where $0 \leq \varepsilon_{\mathrm{f}} < 1$ is an error rate. The error is predominantly relative when $|f(x)|$ is large and effectively absolute when $|f(x)|$ is small, capturing the typical behavior of rounding errors in IEEE 754 floating-point arithmetic [20]. In the numerical experiments, ε_{f} is taken as a large multiple of the machine epsilon to account for accumulated round-off effects. Closely related models are also employed in practical stopping criteria [41].

We next specify the assumptions of gradient evaluations. In practice, termination is typically based on a gradient tolerance $\varepsilon_{\mathrm{gtol}}$, and the condition $\left\|\nabla\overline{f}(x)\right\| \leq \varepsilon_{\mathrm{gtol}}$ is used as a stopping criterion. Hence, prior to termination, it holds that $\left\|\nabla\overline{f}(x)\right\| > \varepsilon_{\mathrm{gtol}}$. If the gradient noise $\left\|\nabla f(x)-\nabla\overline{f}(x)\right\|$ were comparable to or larger than $\varepsilon_{\mathrm{gtol}}$, the stopping criterion could not be satisfied reliably, even when $\nabla f(x)=0$. To ensure meaningful termination, the gradient noise must be sufficiently small relative to $\varepsilon_{\mathrm{gtol}}$. Specifically, during the optimization process, we assume

$$\left\|\nabla f(x)-\nabla\overline{f}(x)\right\| \ll \varepsilon_{\mathrm{gtol}} < \left\|\nabla\overline{f}(x)\right\|.$$

Under this regime, the computed gradient $\nabla\overline{f}(x)$ is a sufficiently accurate approximation of the true gradient $\nabla f(x)$ prior to termination. Accordingly, for the purpose of simplifying the theoretical analysis, we adopt the exact-gradient assumption

$$\nabla\overline{f}(x) = \nabla f(x). \tag{2}$$

We emphasize that this assumption is introduced solely as a modeling simplification, rather than as a claim about practical gradient accuracy. Notably, the proposed algorithm does not rely on Wolfe-type conditions, which typically require accurate gradient information at trial points, and is therefore robust to gradient inaccuracies. Our numerical experiments in section 5 include cases where gradients are contaminated by noise, and they also demonstrate that the proposed method remains effective even when eq. (2) is violated.

2 Proposed Algorithm

We propose a new algorithm for the problem setting stated in section 1.3. We draw inspiration from the OFFO framework to design robust regularization and scaling strategies. Unlike purely OFFO methods, our algorithm exploits function values when they are numerically reliable and smoothly switches to gradient-based control otherwise. This hybrid behavior enables improved stability without sacrificing efficiency in practical optimization tasks. The pseudo-code of our algorithm is given in Algorithm 1. The minimum of the empty set is defined as $+\infty$ in line 3. Algorithm 2 is an auxiliary line search algorithm called in Algorithm 1. In the following, we explain the three main components of Algorithm 1.

Algorithm 1: Noise Tolerant Regularized Quasi-Newton Method

Input: x_0, $0 < k_{\max}$, $0 \leq \varepsilon_{\mathrm{f}} < 1$, $0 < \theta_{\min} \leq \theta_{\max}$ and $0 < \varsigma < 1$

1 $k \leftarrow 0$, $g_0 \leftarrow \nabla \overline{f}(x_0)$, $K^0 \leftarrow \emptyset$, $K^+ \leftarrow \emptyset$
2 **for** $k = 0, 1, 2, \ldots, k_{\max} - 1$ **do**
3 **if** $\min_{j \in K^0} (\overline{f}(x_j) - \Delta_j) \geq \overline{f}(x_k)$ **then**
4 $K^0 \leftarrow K^0 \cup \{k\}$
5 $\mu_k \leftarrow 0$
6 **else**
7 $K^+ \leftarrow K^+ \cup \{k\}$
8 $\mu_k \leftarrow \theta_k \sqrt{\varsigma + \sum_{j \in K^+} \left\| g_j \right\|^2}$ with $\theta_k \in [\theta_{\min}, \theta_{\max}]$ (see section 4.3).
9 $d_k \leftarrow -(B_k + \mu_k I)^{-1} g_k$ with B_k (see section 4.1).
10 Find α_k and Δ_k satisfying eq. (3) by Algorithm 2.
11 $x_{k+1} \leftarrow x_k + \alpha_k d_k$, $g_{k+1} \leftarrow \nabla \overline{f}(x_{k+1})$
12 **end**

Algorithm 2: Backtracking Line Search with Relaxed Armijo Condition

Input: $x_k \in \mathbb{R}^n$, $d_k \in \mathbb{R}^n$, $g_k \in \mathbb{R}^n$, $0 < c < 1$ and $0 < \beta_{\min} < \beta_{\max} < 1$

1 $\alpha_k \leftarrow 1$
2 $\Delta_k \leftarrow \frac{2\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}} \max(1, \overline{f}(x_k), -\overline{f}(x_k + \alpha_k d_k))$
3 **while** $\overline{f}(x_k) + c\alpha_k g_k^\top d_k + \Delta_k < \overline{f}(x_k + \alpha_k d_k)$ **do**
4 $\alpha_k \leftarrow \beta \alpha_k$ with $\beta \in [\beta_{\min}, \beta_{\max}]$ (see section 4.2).
5 $\Delta_k \leftarrow \frac{2\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}} \max(1, \overline{f}(x_k), -\overline{f}(x_k + \alpha_k d_k))$
6 **end**
7 **return** α_k

Computation of the Regularized Quasi-Newton Direction (line 9): The first component is the computation of the regularized quasi-Newton direction d_k stated in section 1.1. Given $g_k = \nabla \overline{f}(x_k) = \nabla f(x_k)$, an approximate Hessian B_k, and a regularization parameter μ_k, we compute the search direction $d_k = -(B_k + \mu_k I)^{-1} g_k$. A practical choice of B_k is discussed in section 4.1.

Relaxed Armijo Condition with an Error-Absorbing Term (line 10): The second component is the computation of the step size α_k using a relaxed Armijo condition, which is defined as follows:

$$\begin{cases} \overline{f}(x_k) + c\alpha_k g_k^\top d_k + \Delta_k \geq \overline{f}(x_k + \alpha_k d_k), \\ \Delta_k = \dfrac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \overline{f}(x_k), -\overline{f}(x_k + \alpha_k d_k)), \end{cases} \tag{3}$$

where Δ_k is the error-absorbing term that is recomputed for each α_k. This relaxation guarantees the existence of $\alpha_k \in (0,1]$ even in the presence of errors in $\overline{f}$. When d_k is a descent direction ($g_k^\top d_k < 0$), eq. (3) ensures that a temporary increase of $\overline{f}$ is at most Δ_k. Since Δ_k depends on $-\overline{f}(x_k + \alpha_k d_k)$ but not on $\overline{f}(x_k + \alpha_k d_k)$, Δ_k does not increase when $\overline{f}(x_k + \alpha_k d_k)$ becomes large. This prevents unbounded growth of Δ_k.

Adaptive Update of the Regularization Parameter (lines 3–8): The third component is the adaptive update of the regularization parameter μ_k. A larger μ_k yields a more conservative step, while a smaller μ_k allows a more aggressive quasi-Newton step, especially when $\mu_k = 0$. In practical settings, $\mu_k = 0$ is preferred, since this allows us to fully exploit the quasi-Newton approximation and typically leads to practically faster convergence. Let us partition the iteration indices $K := \{0, 1, 2, \ldots, k_{\max} - 1\}$ into two sets:

$$K^0 := \{k \in K \mid \mu_k = 0\}, \quad K^+ := \{k \in K \mid \mu_k > 0\}. \tag{4}$$

We set $\mu_k = 0$ if and only if the following condition is satisfied:

$$\min_{j \in K^0,\, j<k} \left(\overline{f}(x_j) - \Delta_j\right) \geq \overline{f}(x_k). \tag{5}$$

Note that $k = 0$ satisfies eq. (5) since K^0 is empty. This condition restricts $\mu_k = 0$ to iterations for which a sufficient decrease in $\overline{f}$ has been observed. Thereby, while $\overline{f}$ can temporarily increase, this condition theoretically ensures that $\overline{f}$ does not grow unboundedly or oscillate indefinitely. Otherwise, to ensure convergence to first-order stationary points, we apply an OFFO-based update

$$\mu_k = \theta_k \sqrt{\varsigma + \sum_{j \in K^+,\, j \leq k} \|g_j\|^2}, \tag{6}$$

as described in section 1.2. Since OFFO schemes do not rely on function values, this update is robust to numerical errors in $\overline{f}$.

3 Theoretical Analysis

In this section, we establish the global convergence properties of Algorithm 1 under the problem setting in section 1.3, namely, eqs. (1) and (2).

3.1 Assumptions

Throughout this section, we make the following three assumptions. The first assumption ensures that the objective function is bounded below.

Assumption 1 *The function f is bounded below, meaning that for all $x \in \mathbb{R}^n$,*

$$f(x) \geq f^* := \inf_{x \in \mathbb{R}^n} f(x) > -\infty.$$

By the triangle inequality and eq. (1), we have $\overline{f}(x) \geq f(x) - \varepsilon_f \max(1, |f(x)|)$. Since the function $g(t) = t - \varepsilon_f \max(1, |t|)$ is non-decreasing for $0 \leq \varepsilon_f < 1$, it follows that $\inf_x g(f(x)) = g(f^*)$. Consequently, we can also lower bound $\overline{f}$ as

$$\overline{f}(x) \geq \overline{f}^* := \inf_{x \in \mathbb{R}^n} \left(f(x) - \varepsilon_f \max(1, |f(x)|)\right) = f^* - \varepsilon_f \max(1, |f^*|) > -\infty. \tag{7}$$

The second assumption ensures smoothness of the objective function.

Assumption 2 *The function $f\colon \mathbb{R}^n \to \mathbb{R}$ is L-smooth, meaning that there exists a constant $L > 0$ such that for all $x, y \in \mathbb{R}^n$,*

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2}\|y - x\|^2.$$

The third assumption requires the matrices $\{B_k\}_{k=0}^{k_{\max}-1}$ to be uniformly positive definite and bounded.

Assumption 3 *There exist constants $0 < m \leq M$ such that all B_k satisfy*

$$mI \preceq B_k \preceq MI.$$

As we will discuss in section 4, the standard BFGS update with a slight modification guarantees Assumption 3 in practice. By Assumption 3, we can derive useful bounds involving g_k and d_k. Using $d_k = -(B_k + \mu_k I)^{-1} g_k$ (line 9) and Assumption 3 asserting $(m + \mu_k)I \preceq B_k + \mu_k I \preceq (M + \mu_k)I$, we obtain the following bounds:

$$-g_k^\top d_k = d_k^\top (B_k + \mu_k I) d_k \geq (m + \mu_k)\|d_k\|^2 \geq m\|d_k\|^2, \tag{8}$$

$$-g_k^\top d_k = g_k^\top (B_k + \mu_k I)^{-1} g_k \geq \frac{1}{M + \mu_k}\|g_k\|^2, \tag{9}$$

$$\|d_k\|^2 = \left\|(B_k + \mu_k I)^{-1} g_k\right\|^2 \leq \frac{1}{(m + \mu_k)^2}\|g_k\|^2. \tag{10}$$

3.2 Error Bound on Function Evaluations

We first establish useful properties of the error-absorbing term Δ_k defined in eq. (3). In the following, we define the actual and observed function decreases as

$$\delta_k \coloneqq f(x_k) - f(x_{k+1}), \tag{11}$$
$$\overline{\delta}_k \coloneqq \overline{f}(x_k) - \overline{f}(x_{k+1}). \tag{12}$$

The first property states that the error between the actual values difference δ_k and the observed values difference $\overline{\delta}_k$ can be bounded by Δ_k.

Lemma 1 *If x_k and x_{k+1} satisfy $f(x_{k+1}) \leq f(x_k)$, it holds that*

$$\delta_k \leq \overline{\delta}_k + \Delta_k.$$

Proof We start from general observations. We first prove that, for all $x \in \mathbb{R}^n$,

$$\begin{cases} \max(1, +f(x)) \leq \frac{1}{1-\varepsilon_f} \max(1, +\overline{f}(x)), \\ \max(1, -f(x)) \leq \frac{1}{1-\varepsilon_f} \max(1, -\overline{f}(x)). \end{cases} \tag{13}$$

We prove the inequality with the plus sign. If $+f(x) \leq 1$, then $\max(1, +f(x)) = 1 \leq \frac{1}{1-\varepsilon_f} \leq \frac{1}{1-\varepsilon_f} \max(1, +\overline{f}(x))$. If $+f(x) > 1$, eq. (1) implies

$$\left| f(x) - \overline{f}(x) \right| \leq \varepsilon_f \max(1, |f(x)|) = +\varepsilon_f f(x) \tag{14}$$

and thus $f(x) - \overline{f}(x) \leq +\varepsilon_f f(x)$. Hence,

$$\max(1, +f(x)) = +f(x) \leq +\frac{1}{1-\varepsilon_f} \overline{f}(x) \leq \frac{1}{1-\varepsilon_f} \max(1, +\overline{f}(x)).$$

The case with the negative sign can be shown analogously, using $-f(x) + \overline{f}(x) \leq -\varepsilon_f f(x)$, which follows from eq. (14). This completes the proof of eq. (13).

We also have the following general bound for all $k \in K$:

$$\begin{aligned} \left| \delta_k - \overline{\delta}_k \right| &= \left| (f(x_k) - \overline{f}(x_k)) - (f(x_{k+1}) - \overline{f}(x_{k+1})) \right| && \text{(rearrangement)} \\ &\leq \left| f(x_k) - \overline{f}(x_k) \right| + \left| f(x_{k+1}) - \overline{f}(x_{k+1}) \right| && \text{(triangle inequality)} \\ &\leq \varepsilon_f \max(1, |f(x_k)|) + \varepsilon_f \max(1, |f(x_{k+1})|) && \text{(eq. (1))} \\ &\leq 2\varepsilon_f \max(1, |f(x_k)|, |f(x_{k+1})|). && \text{(max property)} \end{aligned} \tag{15}$$

With these preparations, we now prove the lemma. From the condition $f(x_{k+1}) \leq f(x_k)$, eq. (15) can be further bounded as

$$\begin{aligned} 2\varepsilon_f \max(1, |f(x_k)|, |f(x_{k+1})|) &= 2\varepsilon_f \max(1, f(x_k), -f(x_{k+1})) && (f(x_{k+1}) \leq f(x_k)) \\ &\leq \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \overline{f}(x_k), -\overline{f}(x_{k+1})) && \text{(eq. (13))} \\ &= \Delta_k, && \text{(eq. (3))} \end{aligned}$$

which yields the desired inequality with $\delta_k - \overline{\delta}_k \leq \left| \delta_k - \overline{\delta}_k \right| \leq \Delta_k$. □

We also derive the following lemma for later use.

Lemma 2 *Under Assumption 3, it holds that*

$$\overline{\delta}_k \le \delta_k + \frac{1+\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}}\Delta_k.$$

Proof Since $-g_k^\top d_k \ge 0$ from Assumption 3 and eq. (8), the relaxed Armijo condition (eq. (3)) implies

$$\overline{f}(x_k) - \overline{f}(x_{k+1}) \ge -c\alpha_k g_k^\top d_k - \Delta_k \ge -\Delta_k. \tag{16}$$

Thus, using the general results in eq. (15) and $\Delta_k \ge 0$, we can further bound as

$$\begin{aligned}
\left|\delta_k - \overline{\delta}_k\right| &\le 2\varepsilon_{\mathrm{f}} \max(1, |f(x_k)|, |f(x_{k+1})|) && \text{(eq. (15))} \\
&\le \frac{2\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}} \max(1, \overline{f}(x_k), -\overline{f}(x_k), \overline{f}(x_{k+1}), -\overline{f}(x_{k+1})) && \text{(eq. (13))} \\
&\le \frac{2\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}} \max(1, \overline{f}(x_k) + \Delta_k, -\overline{f}(x_{k+1}) + \Delta_k) && \text{(eq. (16))} \\
&\le \frac{2\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}} \left(\max(1, \overline{f}(x_k), -\overline{f}(x_{k+1})) + \Delta_k\right) \\
&= \left(1 + \frac{2\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}}\right)\Delta_k = \frac{1+\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}}\Delta_k, && \text{(eq. (3))}
\end{aligned}$$

which yields the desired inequality with $\overline{\delta}_k - \delta_k \le \left|\delta_k - \overline{\delta}_k\right| \le \frac{1+\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}}\Delta_k$. □

3.3 Backtracking Stage

We next analyze the line search in Algorithm 2. Recall that the parameter c in Algorithm 2 satisfies $0 < c < 1$ and $0 < m$ from Assumption 3.

Lemma 3 *Under Assumptions 2 and 3, the backtracking procedure in Algorithm 2 terminates whenever the step size α_k satisfies*

$$\alpha_k \le \frac{2(1-c)m}{L}. \tag{17}$$

Proof Under the L-smoothness of f (Assumption 2), we always have

$$\begin{aligned}
f(x_k) - f(x_k + \alpha_k d_k) &\ge -\alpha_k g_k^\top d_k - \frac{L}{2}\alpha_k^2 \|d_k\|^2 \\
&= -c\alpha_k g_k^\top d_k + \alpha_k \left(-(1-c) g_k^\top d_k - \frac{L\alpha_k}{2}\|d_k\|^2\right).
\end{aligned} \tag{18}$$

From Assumption 3 and eq. (8), we have $-g_k^\top d_k \ge m\|d_k\|^2$. Under eq. (17), we have

$$-(1-c) g_k^\top d_k - \frac{L\alpha_k}{2}\|d_k\|^2 \ge \left((1-c)m - \frac{L\alpha_k}{2}\right)\|d_k\|^2 \ge 0. \tag{19}$$

Combining eqs. (18) and (19) yields

$$f(x_k) - f(x_k + \alpha_k d_k) \geq -c\alpha_k g_k^\top d_k.$$

Since $-g_k^\top d_k \geq 0$ from Assumption 3 and eq. (8), the condition of Lemma 1 is satisfied with $x_{k+1} = x_k + \alpha_k d_k$. Thus we obtain $\delta_k \leq \overline{\delta}_k + \Delta_k$, i.e.,

$$-c\alpha_k g_k^\top d_k \leq f(x_k) - f(x_k + \alpha_k d_k) \leq \overline{f}(x_k) - \overline{f}(x_k + \alpha_k d_k) + \Delta_k.$$

This inequality means that the relaxed Armijo condition (eq. (3)) is satisfied and thus the backtracking procedure terminates. This completes the proof. □

Now, let us analyze Algorithm 2 using Lemma 3. Define

$$\overline{\alpha} := \min\left(\beta_{\min}\frac{2(1-c)m}{L}, 1\right) > 0.$$

The following states the key property of the backtracking procedure.

Proposition 1 *Under Assumptions 2 and 3, the backtracking line search in Algorithm 2 terminates after finitely many rejections, and the step size α_k satisfies*

$$\alpha_k \geq \overline{\alpha}.$$

Proof The line 4 in Algorithm 2 multiplies α_k by β at each rejection, satisfying $0 < \beta_{\min} \leq \beta \leq \beta_{\max} < 1$. For the number of backtracking rejections r, we have $\alpha_k \leq \beta_{\max}^r$. Thus, r is finite since the backtracking procedure terminates for sufficiently small α_k by Lemma 3. If $r = 0$, $\alpha_k = 1 \geq \overline{\alpha}$. Otherwise, $\alpha_k \geq \beta_{\min}\alpha_k'$ where α_k' is the last rejected step size. By Lemma 3, we have $\alpha_k' > \frac{2(1-c)m}{L}$, and thus

$$\alpha_k \geq \beta_{\min}\alpha_k' > \beta_{\min}\frac{2(1-c)m}{L} \geq \overline{\alpha}.$$

This completes the proof. □

3.4 Bound for the case $\mu_k = 0$

In this subsection, we lower bound the sum of actual function decreases δ_k over $k \in K^0$. Recall that K^0 denotes the set of iteration indices k such that $\mu_k = 0$, as in eq. (4). To this end, we establish that the sum of Δ_k over $k \in K^0$ is upper bounded, where Δ_k is the error-absorbing term in the relaxed Armijo condition (eq. (3)). We introduce the following constant $\Delta_{\max}$, which serves as an upper bound for Δ_k of $k \in K^0$.

$$\Delta_{\max} := \frac{2\varepsilon_{\mathrm{f}}}{1-\varepsilon_{\mathrm{f}}}\max(1, \overline{f}(x_0), -\overline{f}^*).$$

Lemma 4 *Under Assumptions 1 to 3, the sum of Δ_k over $k \in K^0$ satisfies*

$$\sum_{k \in K^0} \Delta_k \leq \overline{f}(x_0) - \overline{f}^* + \Delta_{\max}.$$

Proof Let $(k_i)_{i=0}^{\ell}$ with $\ell \geq 0$ be the elements of K^0 in increasing order. We have $\overline{f}(x_{k_\ell}) \leq \overline{f}(x_0)$ from eq. (5) with $k = k_\ell$, where the case $\ell = 0$ follows from $k_0 = 0$. Combining this with Assumption 1, we have $\Delta_{k_\ell} \leq \Delta_{\max}$. For all $i < \ell$, eq. (5) with $j = k_i$ and $k = k_{i+1}$ yields

$$\Delta_{k_i} \leq \overline{f}(x_{k_i}) - \overline{f}(x_{k_{i+1}}), \tag{20}$$

and summing Δ_k over $k \in K^0$ gives

$$\begin{aligned}
\sum_{k \in K^0} \Delta_k &\leq \sum_{i=0}^{\ell-1} (\overline{f}(x_{k_i}) - \overline{f}(x_{k_{i+1}})) + \Delta_{k_\ell} && \text{(eq. (20))} \\
&= \overline{f}(x_0) - \overline{f}(x_{k_\ell}) + \Delta_{k_\ell} && \text{(telescoping sum)} \\
&\leq \overline{f}(x_0) - \overline{f}^* + \Delta_{\max}. && (-\overline{f}(x_{k_\ell}) \leq -\overline{f}^* \text{ and } \Delta_{k_\ell} \leq \Delta_{\max})
\end{aligned}$$

This completes the proof. □

We now bound the sum of $\delta_k = f(x_k) - f(x_{k+1})$ over $k \in K^0$ with $\|g_k\|^2$ as follows.

Lemma 5 *Under Assumptions 1 to 3, the sum of δ_k over $k \in K^0$ satisfies*

$$\sum_{k \in K^0} \delta_k \geq \frac{c\overline{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 - \frac{2}{1 - \varepsilon_{\mathrm{f}}} \left(\overline{f}(x_0) - \overline{f}^* + \Delta_{\max} \right).$$

Proof The relaxed Armijo condition (eq. (3)) at iteration $k \in K^0$ yields

$$\begin{aligned}
\overline{\delta}_k + \Delta_k &= \overline{f}(x_k) - \overline{f}(x_{k+1}) + \Delta_k && \text{(eq. (12))} \\
&\geq -c\alpha_k g_k^\top d_k && \text{(eq. (3))} \\
&\geq \frac{c\overline{\alpha}}{M + \mu_k} \|g_k\|^2 && \text{(Proposition 1 and eq. (9))} \\
&= \frac{c\overline{\alpha}}{M} \|g_k\|^2. && (\mu_k = 0 \text{ since } k \in K^0)
\end{aligned} \tag{21}$$

Using Lemma 2, we also have

$$\overline{\delta}_k + \Delta_k \leq \delta_k + \frac{1 + \varepsilon_{\mathrm{f}}}{1 - \varepsilon_{\mathrm{f}}} \Delta_k + \Delta_k = \delta_k + \frac{2}{1 - \varepsilon_{\mathrm{f}}} \Delta_k. \tag{22}$$

Then, combining eqs. (21) and (22) and summing over $k \in K^0$ gives

$$\sum_{k \in K^0} \frac{c\overline{\alpha}}{M} \|g_k\|^2 \leq \sum_{k \in K^0} \delta_k + \frac{2}{1 - \varepsilon_{\mathrm{f}}} \sum_{k \in K^0} \Delta_k \leq \sum_{k \in K^0} \delta_k + \frac{2}{1 - \varepsilon_{\mathrm{f}}} \left(\overline{f}(x_0) - \overline{f}^* + \Delta_{\max} \right),$$

where the last inequality follows from Lemma 4. Rearranging this completes the proof. □

3.5 Bound for the case $\mu_k > 0$

Next, we show that the sum of δ_k over $k \in K^+$ is lower bounded. Recall that K^+ be a set of iteration indices k such that $\mu_k > 0$ as defined in eq. (4), and that μ_k is defined as $\theta_k \sqrt{\varsigma + \sum_{j \in K^+,\ j<k} \left\|g_j\right\|^2}$ with $\theta_k \in [\theta_{\min}, \theta_{\max}]$ in line 8 of Algorithm 1.

Before proceeding to the main proof, we present Lemma 6, providing standard inequalities in the AdaGrad-Norm algorithm analysis [43, Lemma 3.2], [18, Lemma 3.1]. Its proof is deferred to **??**.

Lemma 6 *In Algorithm 1, we have*

$$\sum_{k \in K^+} \frac{\|g_k\|^2}{\mu_k^2} \leq \frac{1}{\theta_{\min}^2} \left(\ln \left(\varsigma + \sum_{k \in K^+} \|g_k\|^2 \right) - \ln(\varsigma) \right),$$
$$\sum_{k \in K^+} \frac{\|g_k\|^2}{\mu_k} \geq \frac{1}{\theta_{\max}} \left(\sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - \sqrt{\varsigma} \right).$$

We now lower bound the sum of δ_k over $k \in K^+$ with $\|g_k\|^2$. We define the following constants for later use:

$$C_1 := \frac{\overline{\alpha}}{\theta_{\max}}, \qquad C_2 := \frac{M\overline{\alpha} + L/2}{\theta_{\min}^2}.$$

Lemma 7 *Under Assumptions 2 and 3, the sum of δ_k over $k \in K^+$ satisfies*

$$\sum_{k \in K^+} \delta_k \geq C_1 \left(\sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - \sqrt{\varsigma} \right) - C_2 \left(\ln \left(\varsigma + \sum_{k \in K^+} \|g_k\|^2 \right) - \ln(\varsigma) \right).$$

Proof From the L-smoothness (Assumption 2), for $k \in K^+$ we have

$$\begin{aligned}
\delta_k &= f(x_k) - f(x_k + \alpha_k d_k) && \text{(eq. (11) and } x_{k+1} = x_k + \alpha_k d_k) \\
&\geq -\alpha_k g_k^\top d_k - \frac{\alpha_k^2 L}{2} \|d_k\|^2 && \text{(Assumption 2)} \\
&\geq \left(\frac{\alpha_k}{M + \mu_k} - \frac{\alpha_k^2 L}{2(m + \mu_k)^2} \right) \|g_k\|^2 && \text{(eqs. (9) and (10))} \\
&\geq \left(\frac{\overline{\alpha}}{M + \mu_k} - \frac{L}{2(m + \mu_k)^2} \right) \|g_k\|^2. && \text{(Proposition 1 and } \alpha_k \leq 1) \qquad (23)
\end{aligned}$$

We can further bound the terms in eq. (23) as

$$\frac{\overline{\alpha}}{M + \mu_k} = \frac{\overline{\alpha}}{\mu_k} \frac{1}{1 + \frac{M}{\mu_k}} \geq \frac{\overline{\alpha}}{\mu_k} \left(1 - \frac{M}{\mu_k} \right) = \frac{\overline{\alpha}}{\mu_k} - \frac{M\overline{\alpha}}{\mu_k^2}, \quad \frac{L}{2(m + \mu_k)^2} \leq \frac{L}{2\mu_k^2}. \qquad (24)$$

Thus, applying eq. (24) to eq. (23) yields

$$\delta_k \geq \left(\frac{\overline{\alpha}}{\mu_k} - \frac{M\overline{\alpha} + L/2}{\mu_k^2} \right) \|g_k\|^2. \qquad (25)$$

Summing eq. (25) over $k \in K^+$ and applying Lemma 6 yield

$$\sum_{k\in K^+} \delta_k \geq \sum_{k\in K^+} \left(\frac{\overline{\alpha}}{\mu_k} - \frac{M\overline{\alpha} + L/2}{\mu_k^2} \right) \|g_k\|^2$$
$$\geq \frac{\overline{\alpha}}{\theta_{\max}} \left(\sqrt{\varsigma + \sum_{k\in K^+} \|g_k\|^2} - \sqrt{\varsigma} \right) - \frac{M\overline{\alpha} + L/2}{\theta_{\min}^2} \left(\ln\left(\varsigma + \sum_{k\in K^+} \|g_k\|^2 \right) - \ln(\varsigma) \right),$$

which yields the desired result. □

3.6 Global Convergence Rate

Finally, we combine the results for $k \in K^0$ stated in Lemma 5 and $k \in K^+$ stated in Lemma 7 to derive an upper bound on the total sum of $\|g_k\|^2$. We define a constant F as follows:

$$F := f(x_0) - f^* + \frac{2}{1-\varepsilon_{\mathrm{f}}} \left(\overline{f}(x_0) - \overline{f}^* + \Delta_{\max} \right) + C_1 \sqrt{\varsigma} - C_2 \ln(\varsigma). \tag{26}$$

Then, we can obtain the following unified bound of the sum of $\|g_k\|^2$.

Lemma 8 *Under Assumptions 1 to 3, the sum of $\|g_k\|^2$ over $k \in K^0, K^+$ satisfies*

$$F \geq \frac{c\overline{\alpha}}{M} \sum_{k\in K^0} \|g_k\|^2 + C_1 \sqrt{\varsigma + \sum_{k\in K^+} \|g_k\|^2} - C_2 \ln\left(\varsigma + \sum_{k\in K^+} \|g_k\|^2 \right),$$

Proof Since $f(x_{k_{\max}}) \geq f^*$ by Assumption 1 and $K = K^0 \cup K^+$, we have

$$f(x_0) - f^* \geq f(x_0) - f(x_{k_{\max}}) = \sum_{k\in K} \delta_k = \sum_{k\in K^0} \delta_k + \sum_{k\in K^+} \delta_k.$$

From Lemmas 5 and 7, we can further bound as

$$f(x_0) - f^* \geq \frac{c\overline{\alpha}}{M} \sum_{k\in K^0} \|g_k\|^2 - \frac{2}{1-\varepsilon_{\mathrm{f}}} \left(\overline{f}(x_0) - \overline{f}^* + \Delta_{\max} \right)$$
$$+ C_1 \sqrt{\varsigma + \sum_{k\in K^+} \|g_k\|^2} - C_1 \sqrt{\varsigma} - C_2 \ln\left(\varsigma + \sum_{k\in K^+} \|g_k\|^2 \right) + C_2 \ln(\varsigma).$$

Substituting eq. (26) yields the desired bound, concluding the proof. □

Now, we establish the following convergence rate for Algorithm 1.

Theorem 1 *Under Assumptions 1 to 3, the average of squared gradient norms generated by Algorithm 1 satisfies*

$$\frac{1}{k_{\max}} \sum_{k=0}^{k_{\max}-1} \|g_k\|^2 = \mathcal{O}\left(\frac{1}{k_{\max}} \right).$$

Proof Let us define the function $\phi\colon \mathbb{R}_{>0} \to \mathbb{R}$ as

$$\phi(G) \coloneqq \frac{C_1}{2}\sqrt{G} - C_2 \ln(G). \tag{27}$$

Since $\phi'(G) = \frac{C_1}{4\sqrt{G}} - \frac{C_2}{G}$, its minimum over $G > 0$ is attained at $G^* \coloneqq \left(\frac{4C_2}{C_1}\right)^2$ with

$$\phi(G^*) = 2C_2\left(1 - \ln\left(\frac{4C_2}{C_1}\right)\right) = 2C_2 \ln\left(\frac{\mathrm{e}C_1}{4C_2}\right),$$

where e is the base of the natural logarithm. Thus, Lemma 8 yields

$$\begin{aligned} F &\geq \frac{c\overline{\alpha}}{M}\sum_{k\in K^0}\|g_k\|^2 + \frac{C_1}{2}\sqrt{\sum_{k\in K^+}\|g_k\|^2} + \phi\left(\sum_{k\in K^+}\|g_k\|^2 + \varsigma\right) \\ &\geq \frac{c\overline{\alpha}}{M}\sum_{k\in K^0}\|g_k\|^2 + \frac{C_1}{2}\sqrt{\sum_{k\in K^+}\|g_k\|^2} + \phi(G^*), \end{aligned}$$

which is equivalent to

$$0 \leq \frac{c\overline{\alpha}}{M}\sum_{k\in K^0}\|g_k\|^2 + \frac{C_1}{2}\sqrt{\sum_{k\in K^+}\|g_k\|^2} \leq F' \tag{28}$$

where

$$\begin{aligned} F' &\coloneqq F - \phi(G^*) \\ &= f(x_0) - f^* + \frac{2}{1-\varepsilon_{\mathrm{f}}}\left(\overline{f}(x_0) - \overline{f}^* + \Delta_{\max}\right) + C_1\sqrt{\varsigma} - C_2\ln(\varsigma) - 2C_2\ln\left(\frac{\mathrm{e}C_1}{4C_2}\right). \end{aligned}$$

It remains to bound $\sum_{k\in K}\|g_k\|^2 = \sum_{k\in K^0}\|g_k\|^2 + \sum_{k\in K^+}\|g_k\|^2$ given the constraint in eq. (28). More generally, let $a, b > 0$, $c \geq 0$ and $x, y \geq 0$, and consider the maximization problem $x + y$ subject to $ax + b\sqrt{y} \leq c$. The maximum is achieved on the boundary $ax + b\sqrt{y} = c$ since $x + y$ is increasing in both variables. On this boundary, we can write $y = \left(\frac{c-ax}{b}\right)^2$, and thus $x + y$ is convex on an interval $0 \leq x \leq c/a$, attaining its maximum at one of the endpoints. Evaluating at the endpoints yields the candidates $(x, y) = (c/a, 0)$ and $(x, y) = (0, (c/b)^2)$, and therefore the maximum of $x + y$ is $\max\left(c/a, (c/b)^2\right)$. Applying this result to eq. (28) gives

$$\begin{aligned} \frac{1}{k_{\max}}\sum_{k=0}^{k_{\max}-1}\|g_k\|^2 &= \frac{1}{k_{\max}}\left(\sum_{k\in K^0}\|g_k\|^2 + \sum_{k\in K^+}\|g_k\|^2\right) \qquad (K = K^0 \cup K^+) \\ &\leq \frac{1}{k_{\max}}\max\left(\frac{MF'}{c\overline{\alpha}}, \frac{4F'^2}{C_1^2}\right), \end{aligned} \tag{29}$$

which completes the proof. □

As a remark, the function $\phi(G)$ in eq. (27) is closely related to the Lambert W function [7, 18]. For $x \in [-1/\mathrm{e}, 0)$, the equation $y\mathrm{e}^y = x$ admits two real negative solutions, and the smaller one is given by the second branch of Lambert W function $y = W_{-1}(x) < 0$. Using this function, the equation $\phi(G) = 0$ can be solved in closed form, which leads to a sharper bound in Theorem 1. Since this is not essential for the main argument, we omit the details and refer to [18].

We next interpret eq. (29) in the proof of Theorem 1. The contribution of the indices in K^0 is analogous to the classical analysis of gradient descent. Indeed, for gradient descent with fixed step size $\alpha = 1/L$, one has

$$\frac{1}{k_{\max}} \sum_{k=0}^{k_{\max}-1} \|g_k\|^2 \leq \frac{1}{k_{\max}} \frac{2(f(x_0) - f^*)}{\alpha}.$$

Thus, the term associated with K^0 reflects the standard $\mathcal{O}(f(x_0) - f^*)$ dependence of first-order methods since F' contains the initial gaps $f(x_0) - f^*$ and $\overline{f}(x_0) - \overline{f}^*$. In contrast, the contribution of K^+ resembles the behavior of AdaGrad-Norm, whose convergence bound scales quadratically with the initial optimality gap [43].

Finally, Theorem 1 implies that the iteration complexity:

$$\min\left\{k \;\middle|\; \|\nabla f(x_k)\| \leq \varepsilon_{\mathrm{gtol}}\right\} = \mathcal{O}\left(1/\varepsilon_{\mathrm{gtol}}^2\right).$$

This is a standard convergence guarantee for first-order optimization algorithms applied to L-smooth nonconvex functions.

4 Practical Choices of Algorithmic Variables

In sections 2 and 3, we introduced Algorithm 1 and established its convergence properties. In this section, we discuss practical choices of algorithmic variables and parameters that lead to fast and stable performance. The quantities to be discussed here are the matrices B_k, the step sizes α_k, and the regularization parameters μ_k.

4.1 Approximate Hessian B_k

We first discuss the construction of the matrices B_k used in line 9 of Algorithm 1. The purpose of this subsection is to show that B_k can satisfy Assumption 3, the uniform spectral bound, with a slight modification and selection of curvature pairs using the L-BFGS method.

Construction of B_k: We briefly note the L-BFGS method and the curvature pair notation [4, 25, 31]. For an iteration k, we define

$$s_k \coloneqq x_{k+1} - x_k, \qquad y_k \coloneqq g_{k+1} - g_k$$

where g_k and g_{k+1} are the gradients at x_k and x_{k+1}, respectively. Starting from an initial matrix, an approximate Hessian is obtained by successively applying BFGS

updates to a sequence of curvature pairs. Let $\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$ be a collection of curvature pairs, where k_i denotes the iteration index at which the i-th pair was generated. Starting from the initial matrix $B^{(0)} = \gamma I$ with $\gamma = \|y_{k_0}\|^2 / y_{k_0}^\top s_{k_0}$, we define $\mathrm{BFGS}(\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1})$ as the matrix $B^{(p)}$ obtained by sequentially applying the BFGS update as

$$B^{(j+1)} = B^{(j)} - \frac{B^{(j)} s_{k_j} s_{k_j}^\top B^{(j)}}{s_{k_j}^\top B^{(j)} s_{k_j}} + \frac{y_{k_j} y_{k_j}^\top}{y_{k_j}^\top s_{k_j}}$$

for $j = 0, 1, \ldots, p-1$, where each curvature pair (s_{k_j}, y_{k_j}) is applied in order. Here, $p > 0$ denotes the maximum number of stored curvature pairs and is a fixed small integer in the L-BFGS method. When fewer than p curvature pairs are available, the BFGS update is applied using all currently available pairs.

Now, we provide a sufficient condition for Assumption 3 by the following proposition, which is a variant of existing work [12, Theorem 5.5] [14, Lemma 1]. Its proof is deferred to **??**.

Proposition 2 *Let $\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$ be a fixed number of curvature pairs with $s_{k_i} \neq 0$. Assume that there exist constants $0 < \lambda \leq \Lambda$ such that for all $(s, y) \in \{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$,*

$$y^\top s \geq \frac{1}{\Lambda} \|y\|^2, \tag{30}$$

$$y^\top s \geq \lambda \|s\|^2. \tag{31}$$

Define the condition number $\kappa := \Lambda / \lambda$. Then there exist $0 < m < M$ such that

$$mI \preceq \mathrm{BFGS}(\{s_{k_i}, y_{k_i}\}_{i=0}^{p-1}) \preceq MI, \tag{32}$$

where

$$M := (1+p)\Lambda, \qquad m := \left((1+\sqrt{\kappa})^{2p} \left(\frac{1}{\lambda} + \frac{1}{\lambda(2\sqrt{\kappa} + \kappa)} \right) \right)^{-1}.$$

Proposition 2 shows that, whenever eqs. (30) and (31) hold for the stored pairs, the resulting BFGS matrix satisfies Assumption 3 as expressed in eq. (32).

Based on Proposition 2, we basically adopt the following rule for forming B_k:

$$B_k = \mathrm{BFGS}(\{(s_{k_i}, \bar{y}_{k_i})\}), \tag{33}$$

where

$$\bar{y}_{k_i} := \theta_i^{\mathrm{damp}} y_{k_i} + (1 - \theta_i^{\mathrm{damp}}) B_{k-1} s_{k_i}$$

with $\theta_i^{\mathrm{damp}} \in [0, 1]$. This parameter is determined using a damped BFGS update [26, 33] and the indices $\{k_i\}_{i=0}^{p-1}$ are selected so that the modified pairs $(s_{k_i}, \bar{y}_{k_i})$ satisfy eqs. (30) and (31) for some constants λ, Λ. The damped BFGS is a classical technique to ensure $\bar{y}_{k_i}^\top s_{k_i} > 0$, and we adopted its simple limited-memory variant [26].

Since we do not employ Wolfe conditions in the line search, this damping technique plays a central role in ensuring the positive definiteness of B_k. The selection of curvature pairs shares the same spirit as the cautious update [23, 27, 28], which only uses pairs satisfying certain curvature conditions. By Proposition 2, B_k satisfies the uniform spectral bound in Assumption 3. This confirms that the practical construction of B_k is consistent with the theoretical requirements.

Remarks for Practical Implementation: Having established that the proposed construction of B_k satisfies Assumption 3, we now present several remarks on its practical implementation. We begin by describing how to compute the search direction

$$d_k = -(B_k + \mu_k I)^{-1} g_k$$

with B_k in eq. (33). When $\mu_k = 0$, the direction can be computed efficiently by the standard L-BFGS two-loop recursion [25, 31]. For the case $\mu_k > 0$, regularized limited-memory schemes may in principle be employed [22]. Alternatively, one may adopt the simple and robust approximation mentioned in the same reference. Specifically, we approximate B_k in eq. (33) by

$$B_k \approx \mathrm{BFGS}(\{s_{k_i}, \bar{y}_{k_i} + \mu_k s_{k_i}\}_{i=0}^{p-1}) - \mu_k I, \tag{34}$$

so that $B_k + \mu_k I \approx \mathrm{BFGS}(\{s_{k_i}, \bar{y}_{k_i} + \mu_k s_{k_i}\}_{i=0}^{p-1})$. Its inverse can then be computed using the standard two-loop recursion. Although this approximation is inexact, it has been reported to yield nearly identical practical performance while remaining comparatively simple to implement [22]. We adopt this approximation in our numerical experiments.

Separately, there exist function-based modified secant conditions [1, 19, 45, 48] that adjust the secant equation using available function values when those quantities are reliable. Such adjustments aim to improve practical convergence behavior. We partially adopt these function-based modifications following prior work [46, 49]. The suffix `-MS` in section 5 indicates the use of this technique. Because these modifications are well established, we omit low-level implementation details here. Further specifics are available in the cited references and in our open-source implementation.

4.2 Step Size α_k

We next discuss the adjustment of the step size α_k in line 4 of Algorithm 2. As indicated in the pseudo-code, we choose α_k using interpolation of the objective function. Moré–Thuente line searches [29] are widely used in quasi-Newton methods [41], and they typically employ cubic or quadratic interpolation to select trial step sizes. In our implementation, we use an essentially identical procedure with a slight modification, namely, clipping the trial step size to the interval $[\beta_{\min}\alpha_k, \beta_{\max}\alpha_k]$. For all numerical experiments, we set the Armijo parameter to $c = 10^{-4}$ and choose the interpolation bounds as $\beta_{\min} = 1/16$ and $\beta_{\max} = 15/16$.

Furthermore, when $\mu_k > 0$ and $\alpha_k = 1$, we introduce a one-time adjustment of the trial step size using gradient information. Let d_k be the search direction and g_k and g_{try} the gradients at the current and trial points. If the directional derivative along d_k changes sign, $d_k^\top g_k < 0$ and $d_k^\top g_{\mathrm{try}} > 0$, and the trial gradient is sufficiently aligned with d_k, namely $d_k^\top g_{\mathrm{try}} > 0.5\|d_k\|\|g_{\mathrm{try}}\|$, we rescale α_k once by a secant-like factor,

$$\alpha_k \leftarrow \mathrm{clip}\left(\alpha_k \frac{-d_k^\top g_k}{d_k^\top g_{\mathrm{try}} - d_k^\top g_k}, \beta_{\min}\alpha_k, \beta_{\max}\alpha_k\right),$$

where $\mathrm{clip}(x,a,b) \coloneqq \min(\max(x,a),b)$ is a clipping function. When objective values are unreliable and backtracking and interpolation do not occur, this correction effectively refines the step size. Since this adjustment is at most once per iteration, the theoretical results in section 3 remain valid.

4.3 Regularization Parameter μ_k

We finally discuss the adjustment of the regularization parameter μ_k in line 8 of Algorithm 1. In our implementation, we choose the regularization parameter as

$$\mu_k = \mathrm{clip}\left(\frac{1}{10}\|g_k\|, \frac{1}{100}G_k, G_k\right) \quad \text{where} \quad G_k \coloneqq \sqrt{\varsigma + \sum_{j\in K,\, j\le k} \|g_j\|^2},$$

which is consistent with the general form stated in eq. (6) with an adaptive factor $\theta_k \in [10^{-2}, 1]$. In practice, we may set $\varsigma = 0$ as long as the initial gradient is nonzero since ς is mainly introduced for avoiding zero division, but to make it consistent with the theory, we set $\varsigma = 10^{-10}$ in all experiments.

When B_k is a good approximation of the Hessian, the regularization parameter μ_k should be small for fast convergence. Since the quantity G_k is strictly increasing with k, it is natural in practice to decrease the effective regularization when sufficient progress in the objective function is achieved. When the difference between the left- and right-hand sides of eq. (5) is larger than 1, we restart the accumulation by resetting K^+ to the empty set and G_k to ς. As long as the number of such restarts is finite, the overall convergence analysis in section 3 remains valid. Since these choices are heuristic, there may be room for further improvement. Especially, further increasing μ_k stabilizes the method when the objective function values are highly noisy, and decreasing μ_k more aggressively may accelerate convergence when the objective function is accurate.

5 Numerical Experiments

We conducted numerical experiments to evaluate the practical performance of our proposed method and to compare it with existing optimization algorithms.

5.1 Methods

We compared the following methods in our experiments.

- (`Ours`): Our proposed regularized quasi-Newton method
  - We used Algorithms 1 and 2 with the details in section 4.
  - (`Ours-MS`) means the variant incorporating the modified secant condition described in section 4.1.
- (`Line`): Line search L-BFGS method
  - Our Python implementation ported from LBFGSpp [32].

  - The step size was determined using the Moré–Thuente line search, closely resembling SciPy's implementation.
  - (`Line-MS`) means the same variant as `Ours-MS`.
- (`Reg`): Regularized L-BFGS method [22]
  - A quadratic regularization-based quasi-Newton method that does not rely on line search and is conceptually closer to trust-region methods.
  - We wrapped the function `regLBFGS` with `solveNonmonotone` from [22].
  - (`Reg-Sec`) means the incorporation of the approximated computation as in eq. (34). This is a wrapped function of `regLBFGSsec` from [22].
- (`SciPy`): SciPy's L-BFGS-B method [41]
  - An established L-BFGS method implemented in C and called from Python.
  - We used the default parameters, with the exception that `ftol` was set to 0 to avoid premature termination.
- (`NTQN`): Noise-tolerant quasi-Newton method [37, 44]
  - A method designed to accommodate inexact gradient information, as mentioned in section 1.3.
  - We mainly used the default parameter settings, and slightly modified the stopping criteria to terminate at the desired gradient norm tolerance.

In the following, each method is referred to by the name given in parentheses. All these methods are L-BFGS-based, and we set the number of stored curvature pairs to 10, the default value in SciPy's L-BFGS-B implementation. All experiments were conducted on a Microsoft Windows 11 Home system (version 10.0, build 26100) equipped with a 13th Gen Intel® Core™ i7-1360P CPU, featuring 12 physical cores and 16 logical processors. All computations were performed using the CPU only.

5.2 Experimental Results

5.2.1 Test Problems and Settings

We evaluated all algorithms on unconstrained problems from the CUTEst benchmark collection [13], implemented in Python using the PyCUTEst interface [11]. All problems were initialized with the default starting points provided by the library. Following the previous work [22], we excluded problems such as the initial point being already stationary or containing `NaN` and `INF` in its function values or gradient. To simulate different numerical environments, we tested the following four settings:

- Artificially noised 64-bit floating-point precision (220 problems, $\varepsilon_{\mathrm{f}} = 10^{-2}$)
- 64-bit floating-point precision (220 problems, $\varepsilon_{\mathrm{f}} = 2.22 \times 10^{-9}$)
- 32-bit floating-point precision (220 problems, $\varepsilon_{\mathrm{f}} = 1.19 \times 10^{-3}$)
- 16-bit floating-point precision (152 problems, $\varepsilon_{\mathrm{f}} = 9.77 \times 10^{-2}$)

In the artificial noise setting, we added uniform random noise to both the objective function values and each component of the gradient vectors independently, i.e.,

$$\overline{f}(x) = f(x) + \mathrm{unif}(-10^{-3}, 10^{-3}), \quad \nabla\overline{f}(x) = \nabla f(x) + \mathrm{unif}(-10^{-3}, 10^{-3})\mathbb{1},$$

explanation	min abs value	relative error	ε_f
64-bit double precision	2.23×10^{-308}	$2^{-52} \approx 2.22 \times 10^{-16}$	2.22×10^{-9}
32-bit single precision	1.18×10^{-38}	$2^{-23} \approx 1.19 \times 10^{-7}$	1.19×10^{-3}
16-bit half precision	6.10×10^{-5}	$2^{-10} \approx 9.77 \times 10^{-4}$	9.77×10^{-2}

Table 1: The "min abs value" is the smallest positive normalized number, the "relative error" is the maximum relative rounding error, and ε_f is the parameter in (1) used in our experiments.

where $\text{unif}(a,b)$ denotes a uniform random variable in the interval $[a,b]$ and $\mathbb{1}$ is the vector of all ones. This setting is the same as the one in [37]. In the reduced precision settings, to simulate lower numerical precision, we first cast the input vector x to the target lower precision, then computed $\overline{f}(x)$ and $\nabla\overline{f}(x)$ using the lower-precision input. Although the internal computations of CUTEst remain in 64-bit precision, the casting of the input vector effectively simulated the impact of reduced precision on function and gradient evaluations.

Each setting uses a different error rate ε_f in the error model (eq. (1)), as summarized in Table 1. The table relates the machine epsilon (maximum relative rounding error) for each floating-point precision to the ε_f value used in our experiments. The relatively large ε_f values are chosen to account for the cumulative errors that may arise during optimization. We also set the maximum number of iterations to $k_{\max} = 15{,}000$ and timed out runs exceeding 10 minutes per problem.

5.2.2 Performance Profiles

We used performance profiles [9] to compare the algorithms. Let P denote the set of test problems and S the set of solvers. For each problem $p \in P$ and solver $s \in S$, let $n_{p,s} > 0$ denote the number of oracle calls (i.e., function and gradient evaluations) required to reach a specified gradient norm tolerance $\varepsilon_{\text{gtol}}$. We use the number of oracle calls rather than clock time to avoid the influence of algorithmic overhead and to focus on the intrinsic optimization efficiency. When a solver fails to converge within the gradient tolerance or exceeds a maximum number of oracle calls, we set $n_{p,s} = +\infty$. Then, the performance ratio $r_{p,s}$ is defined as

$$r_{p,s} = \frac{n_{p,s}}{\min_{s' \in S} n_{p,s'}}.$$

For a given threshold $\tau \geq 1$ and solver s, the performance profile plots the proportion of problems for which $r_{p,s} \leq \tau$. This means that the y-coordinate for solver s at x-coordinate τ is given by

$$\rho_s(\tau) = \frac{1}{|P|} \left| \{ p \in P \mid r_{p,s} \leq \tau \} \right|.$$

A curve that lies above others indicates better performance. Following standard implementations [41], we used the infinity norm for the gradient, though other norms may equally be employed.

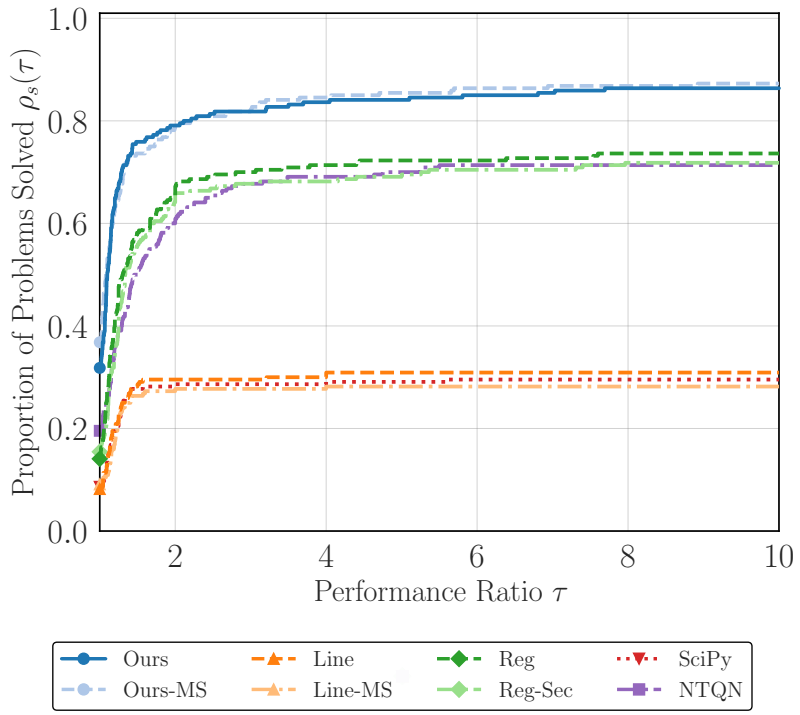


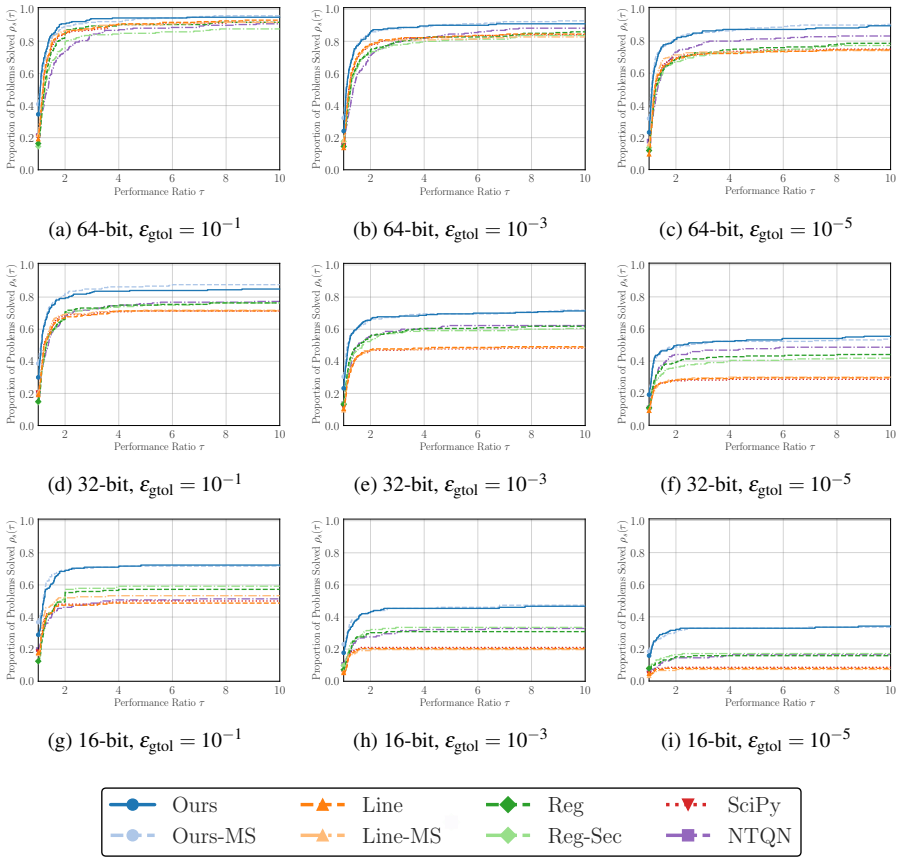
Fig. 1: Performance profile for artificial noise setting (level of 10^{-3}) and $\varepsilon_{\text{gtol}} = 10^{-2}$.

5.2.3 Results with Artificial Noise

We first present the result in the artificial noise setting described in section 5.2.1. Figure 1 presents the performance profile. In this regime, the proposed methods show superior robustness compared to other quasi-Newton methods. This result highlights the validity of our theoretical analysis in section 3 and demonstrates the effectiveness of our approach in handling noisy objective function values.

5.2.4 Results at Different Precision Levels

We next present performance profiles for each floating-point precision (64-, 32-, and 16-bit). We evaluated each method at three gradient norm tolerance levels, i.e., $\varepsilon_{\text{gtol}} \in \{10^{-1}, 10^{-3}, 10^{-5}\}$. These tolerance levels represent different optimization regimes, from moderate accuracy to high precision requirements. Figure 2 shows the comprehensive performance profiles across all three precision levels. Our methods (`Ours`) and (`Ours-MS`) demonstrate competitive or superior performance compared to existing methods in the standard 64-bit setting and maintain robust performance as precision decreases to 32-bit and 16-bit. The proposed methods maintain stability



(a) 64-bit, $\varepsilon_{\text{gtol}} = 10^{-1}$ (b) 64-bit, $\varepsilon_{\text{gtol}} = 10^{-3}$ (c) 64-bit, $\varepsilon_{\text{gtol}} = 10^{-5}$

(d) 32-bit, $\varepsilon_{\text{gtol}} = 10^{-1}$ (e) 32-bit, $\varepsilon_{\text{gtol}} = 10^{-3}$ (f) 32-bit, $\varepsilon_{\text{gtol}} = 10^{-5}$

(g) 16-bit, $\varepsilon_{\text{gtol}} = 10^{-1}$ (h) 16-bit, $\varepsilon_{\text{gtol}} = 10^{-3}$ (i) 16-bit, $\varepsilon_{\text{gtol}} = 10^{-5}$

Fig. 2: Performance profiles across different floating-point precisions (64-, 32-, and 16-bit) and tolerance levels. The results demonstrate the robustness of the proposed methods.

even when traditional line search methods encounter difficulties with reduced precision arithmetic.

5.3 Computational Time per Iteration

In this subsection, we present an experiment to compare the per-iteration computational cost of each method. Here, our purpose lies in the measurement of the total execution time rather than the number of oracle calls. We conducted experiments on the ill-conditioned quadratic problem

$$\underset{x\in\mathbb{R}^n}{\text{minimize}} \quad \frac{1}{2}\sum_{i=1}^{n} i x_i^2$$

Fig. 3: Mean execution times for the experiment in section 5.3. This plot roughly indicates the extra computational overhead of each method. Error bars represent one standard deviation over 100 runs.

with $n = 10{,}000$. The initial point x_0 was chosen as the all-ones vector. We used memory size $m = 10$ and maximum iterations $k_{\max} = 100$, and we did not set any stopping criteria other than reaching $k_{\max}$. The simplicity of the optimization problem and the small value of $k_{\max}$ are well suited to the purpose of this experiment, as they ensure that the total runtime is dominated by algorithmic overhead rather than function or gradient computation. Timing data were collected after three warm-up runs, followed by 100 recorded trials.

In Figure 3, we report the execution time statistics, and we can observe that our proposed methods (`Ours`) and (`Ours-MS`) exhibit competitive performance compared to other methods. Combining with the results from section 5.2, our methods are competitive or superior in total oracle calls and per-iteration computational cost, indicating practical efficiency. As a side note, although the (`SciPy`) calls optimized C routines, it incurs associated dispatch overhead, producing a larger mean time. One of the reasons why the (`Reg`) and (`Reg-Sec`) methods take longer execution time is due to the implementation aspect of the original code.

In Table 2, we report the total number of oracle calls (calls of $\overline{f}$ and $\nabla\overline{f}$) and final objective values $\overline{f}(x_{k_{\max}})$. The purpose of this table is to confirm that all methods achieved similar convergence behavior within the limited number of iterations. The comparable values in the table indicate that the comparison of execution times is fair and not affected by differences in convergence behavior.

	Line	Ours	Line-MS	Ours-MS	NTQN	SciPy	Reg-Sec	Reg
# Calls	212	202	212	202	219	212	206	206
$\overline{f}(x_{k_{\max}})$	1.24	1.34	1.24	1.34	1.43	1.24	1.48	1.47

Table 2: The number of oracle calls and final observed objective values for the experiment in section 5.3.

6 Conclusion

In this work, we proposed a regularized quasi-Newton method for noisy nonconvex optimization and established its global convergence properties. We also demonstrated the method's effectiveness through numerical experiments on standard benchmark problems under various noise levels and precision settings. The results indicate that our methods are robust and effective, particularly in noisy or low-precision environments where traditional line search methods may struggle.

Future work includes the investigation of local convergence properties, the extension to constrained optimization, and the applications to practical problems in machine learning and scientific computing. Additionally, exploring adaptive strategies for selecting algorithmic parameters based on problem characteristics could further enhance the method's performance. An explicit analysis of the effect of inexact gradient evaluations would also provide valuable insight into the method's robustness and guide further enhancements.

References

1. Babaie-Kafaki S, Ghanbari R, Mahdavi-Amiri N (2010) Two new conjugate gradient methods based on modified secant equations. Journal of Computational and Applied Mathematics 234(5):1374–1386, DOI 10.1016/j.cam.2010.01.052
2. Bellavia S, Gurioli G, Morini B, Toint PL (2021) The Impact of Noise on Evaluation Complexity: The Deterministic Trust-Region Case. DOI 10.48550/arXiv.2104.02519, 2104.02519
3. Berahas AS, Cao L, Choromanski K, Scheinberg K (2021) A Theoretical and Empirical Comparison of Gradient Approximations in Derivative-Free Optimization. DOI 10.48550/arXiv.1905.01332, 1905.01332
4. Berahas AS, Curtis FE, Zhou B (2022) Limited-memory BFGS with displacement aggregation. Mathematical Programming 194(1):121–157, DOI 10.1007/s10107-021-01621-6
5. Carter RG (1993) Numerical Experience with a Class of Algorithms for Nonlinear Optimization Using Inexact Function and Gradient Information. SIAM Journal on Scientific Computing 14(2):368–388, DOI 10.1137/0914023
6. Clancy RJ, Menickelly M, Hückelheim J, Hovland P, Nalluri P, Gjini R (2022) TROPHY: Trust Region Optimization Using a Precision Hierarchy. In: Computational Science – ICCS 2022, Springer, Cham, pp 445–459, DOI 10.1007/978-3-031-08751-6_32

7. Corless RM, Gonnet GH, Hare DEG, Jeffrey DJ, Knuth DE (1996) On the LambertW function. Advances in Computational Mathematics 5(1):329–359, DOI 10.1007/BF02124750
8. Doikov N, Mishchenko K, Nesterov Y (2024) Super-Universal Regularized Newton Method. SIAM Journal on Optimization 34(1):27–56, DOI 10.1137/22M1519444
9. Dolan ED, Moré JJ (2002) Benchmarking optimization software with performance profiles. Mathematical Programming 91(2):201–213, DOI 10.1007/s101070100263
10. Duchi J, Hazan E, Singer Y (2011) Adaptive Subgradient Methods for Online Learning and Stochastic Optimization. Journal of Machine Learning Research 12(61):2121–2159
11. Fowkes J, Roberts L, Bűrmen Á (2022) PyCUTEst: An open source Python package of optimization test problems. Journal of Open Source Software 7(78):4377, DOI 10.21105/joss.04377
12. Gao W, Goldfarb D (2018) Quasi-Newton Methods: Superlinear Convergence Without Line Searches for Self-Concordant Functions. DOI 10.48550/arXiv.1612.06965, 1612.06965
13. Gould NI, Orban D, Toint PL (2015) CUTEst: A Constrained and Unconstrained Testing Environment with safe threads for mathematical optimization. Comput Optim Appl 60(3):545–557, DOI 10.1007/s10589-014-9687-3
14. Gower RM, Goldfarb D, Richtárik P (2016) Stochastic Block BFGS: Squeezing More Curvature out of Data. DOI 10.48550/arXiv.1603.09649, 1603.09649
15. Grapiglia GN, Stella GFD (2022) An adaptive trust-region method without function evaluations. Computational Optimization and Applications 82(1):31–60, DOI 10.1007/s10589-022-00356-0
16. Gratton S, Toint PL (2020) A note on solving nonlinear optimization problems in variable precision. Computational Optimization and Applications 76(3):917–933, DOI 10.1007/s10589-020-00190-2
17. Gratton S, Toint PL (2023) OFFO minimization algorithms for second-order optimality and their complexity. Computational Optimization and Applications 84(2):573–607, DOI 10.1007/s10589-022-00435-2
18. Gratton S, Jerad S, Toint PhL (2024) Complexity of a class of first-order objective-function-free optimization algorithms. Optimization Methods and Software 0(0):1–31, DOI 10.1080/10556788.2023.2296431
19. Hassan BA, Moghrabi IAR (2023) A modified secant equation quasi-Newton method for unconstrained optimization. Journal of Applied Mathematics and Computing 69(1):451–464, DOI 10.1007/s12190-022-01750-x
20. Higham NJ (2002) Accuracy and Stability of Numerical Algorithms. Society for Industrial and Applied Mathematics, DOI 10.1137/1.9780898718027
21. Izycheva A, Darulova E (2017) On sound relative error bounds for floating-point arithmetic. In: Proceedings of the 17th Conference on Formal Methods in Computer-Aided Design, FMCAD Inc, Austin, Texas
22. Kanzow C, Steck D (2023) Regularization of limited memory quasi-Newton methods for large-scale nonconvex minimization. Mathematical Programming Computation 15(3):417–444, DOI 10.1007/s12532-023-00238-4

23. Li DH, Fukushima M (2001) On the Global Convergence of the BFGS Method for Nonconvex Unconstrained Optimization Problems. SIAM Journal on Optimization 11(4):1054–1064, DOI 10.1137/S1052623499354242
24. Li YJ, Li DH (2009) Truncated regularized Newton method for convex minimizations. Computational Optimization and Applications 43(1):119–131, DOI 10.1007/s10589-007-9128-7
25. Liu DC, Nocedal J (1989) On the limited memory BFGS method for large scale optimization. Mathematical Programming 45(1):503–528, DOI 10.1007/BF01589116
26. Lotfi S, de Ruisselet TB, Orban D, Lodi A (2020) Stochastic Damped L-BFGS with Controlled Norm of the Hessian Approximation. DOI 10.13140/RG.2.2.27851.41765/1, 2012.05783
27. Mannel F (2025) A Globalization of L-BFGS and the Barzilai–Borwein Method for Nonconvex Unconstrained Optimization. Journal of Optimization Theory and Applications 204(3):1–34, DOI 10.1007/s10957-024-02565-5
28. Mannel F, Om Aggrawal H, Modersitzki J (2024) A structured L-BFGS method and its application to inverse problems. Inverse Problems 40(4):045022, DOI 10.1088/1361-6420/ad2c31
29. Moré JJ, Thuente DJ (1994) Line search algorithms with guaranteed sufficient decrease. ACM Trans Math Softw 20(3):286–307, DOI 10.1145/192115.192132
30. Nesterov Y, Polyak B (2006) Cubic regularization of Newton method and its global performance. Mathematical Programming 108(1):177–205, DOI 10.1007/s10107-006-0706-8
31. Nocedal J, Wright SJ (1999) Numerical Optimization. Springer, DOI 10.1007/978-0-387-40065-5
32. Okazaki N (2007) libLBFGS: A library of limited-memory broyden-fletcher-goldfarb-shanno (l-BFGS)
33. Powell MJ (1978) Algorithms for nonlinear constraints that use lagrangian functions. Math Program 14(1):224–248, DOI 10.1007/BF01588967
34. Ranganath A, Singhal M, Marcia R (2025) Symmetric Rank-One Quasi-Newton Methods for Deep Learning Using Cubic Regularization. DOI 10.48550/arXiv.2502.12298, 2502.12298
35. Ruan H, Yang W (2024) Adaptive Regularized Quasi-Newton Method Using Inexact First-Order Information. Journal of Computational Mathematics 42(6):1656–1687, DOI 10.4208/jcm.2306-m2022-0279
36. Sheng Z, Yuan G, Cui Z (2018) A new adaptive trust region algorithm for optimization problems. Acta Mathematica Scientia 38(2):479–496, DOI 10.1016/S0252-9602(18)30762-8
37. Shi HJM, Xie Y, Byrd R, Nocedal J (2022) A Noise-Tolerant Quasi-Newton Algorithm for Unconstrained Optimization. SIAM Journal on Optimization 32(1):29–55, DOI 10.1137/20M1373190
38. Sun S, Nocedal J (2023) A trust region method for noisy unconstrained optimization. Mathematical Programming 202(1):445–472, DOI 10.1007/s10107-023-01941-9
39. Ueda K, Yamashita N (2010) Convergence Properties of the Regularized Newton Method for the Unconstrained Nonconvex Optimization. Applied Mathematics

and Optimization 62(1):27–46, DOI 10.1007/s00245-009-9094-9
40. Ueda K, Yamashita N (2014) A regularized Newton method without line search for unconstrained optimization. Computational Optimization and Applications 59(1):321–351, DOI 10.1007/s10589-014-9656-x
41. Virtanen P, Gommers R, Oliphant TE, Haberland M, Reddy T, Cournapeau D, Burovski E, Peterson P, Weckesser W, Bright J, van der Walt SJ, Brett M, Wilson J, Millman KJ, Mayorov N, Nelson ARJ, Jones E, Kern R, Larson E, Carey CJ, Polat I, Feng Y, Moore EW, VanderPlas J, Laxalde D, Perktold J, Cimrman R, Henriksen I, Quintero EA, Harris CR, Archibald AM, Ribeiro AH, Pedregosa F, van Mulbregt P, SciPy 10 Contributors (2020) SciPy 1.0: Fundamental algorithms for scientific computing in python. Nature Methods 17:261–272, DOI 10.1038/s41592-019-0686-2
42. Wang S, Fadili J, Ochs P (2025) Convergence rates of regularized quasi-Newton methods without strong convexity. DOI 10.48550/arXiv.2506.00521, 2506.00521
43. Ward R, Wu X, Bottou L (2020) AdaGrad stepsizes: Sharp convergence over nonconvex landscapes. J Mach Learn Res 21(1):219:9047–219:9076
44. Xie Y, Byrd RH, Nocedal J (2020) Analysis of the BFGS method with errors. SIAM Journal on Optimization 30(1):182–209
45. Yabe H, Ogasawara H, Yoshino M (2007) Local and superlinear convergence of quasi-Newton methods based on modified secant conditions. Journal of Computational and Applied Mathematics 205(1):617–632, DOI 10.1016/j.cam.2006.05.018
46. Yuan G, Wei Z (2010) Convergence analysis of a modified BFGS method on convex minimizations. Computational Optimization and Applications 47(2):237–255, DOI 10.1007/s10589-008-9219-0
47. Zhang H, Ni Q (2018) A new regularized quasi-Newton method for unconstrained optimization. Optimization Letters 12(7):1639–1658, DOI 10.1007/s11590-018-1236-z
48. Zhang J, Xu C (2001) Properties and numerical performance of quasi-Newton methods with modified quasi-Newton equations. Journal of Computational and Applied Mathematics 137(2):269–278, DOI 10.1016/S0377-0427(00)00713-5
49. Zhang JZ, Deng NY, Chen LH (1999) New Quasi-Newton Equation and Related Methods for Unconstrained Optimization. Journal of Optimization Theory and Applications 102(1):147–167, DOI 10.1023/A:1021898630001